

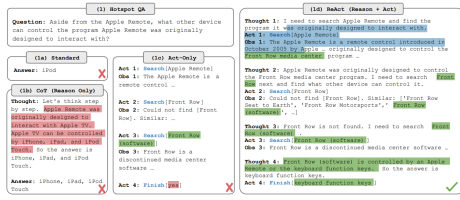
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agent-spec-kit

Scenario-Driven Evaluation and Testing of Tool-Using LLM Agents

Tool-using agents act on the world

- LLMs are no longer just text generators — as **agents** they plan, call tools, and **change external state**.
- Correctness can therefore **not** be judged from the final reply alone.
- A support agent can sound perfectly polite while it:
 - skips **authentication** and leaks account data,
 - calls the **wrong tool** or in the wrong order,
 - leaves the **database** in an invalid state.



Reasoning + acting (ReAct): the agent interleaves thoughts, tool calls, and replies.

The plausible answer is only one part of being correct.

Correctness lives on three surfaces

Output is necessary but far from sufficient



Output

what the agent says



Trace

which tools it called



State

the state it left behind

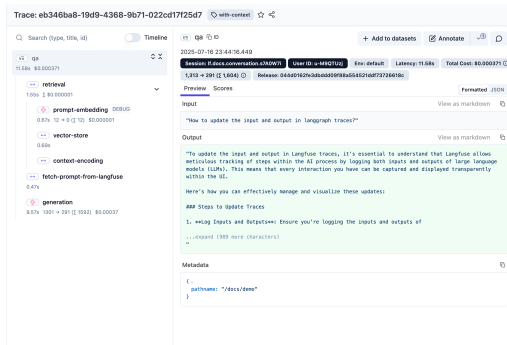
A failure on any surface can be hidden by a fluent reply — and may even be masked by recovery on a later turn.

Existing tools only cover part of the loop

- **Output graders / LLM-as-judge** (HELM, G-Eval, MT-Bench): great for comparing models on isolated final answers — but blind to tool and state failures.
- **Agent benchmarks** (τ -bench, AppWorld, WebArena): trace/state aware, but fixed leaderboards, not reusable infrastructure for your own agent.
- **Trace viewers** (LangSmith, Langfuse): show what happened — the developer still has to infer what should have happened.

The gap

No single, reusable way to assert over **output + trace + state** in one multi-turn scenario — with **precise diagnostics** when a check fails.



Trace viewers show the episode; they do not encode the expected behaviour.

Research question

Can the evaluation of **stateful, tool-using LLM agents** be expressed in a **software-testing style** —

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readable scenarios that combine assertions over **replies**, **tool trajectories**, and **application state**, while preserving enough structured evidence to **diagnose** failures precisely?

Answer: agent-spec-kit — a scenario-driven testing framework for LLM agents, evaluated on a 50-task benchmark.

Part I --- The Library

agent-spec-kit: behaviour as executable scenarios

The core idea: a scenario is a conversation script

Output, tool, and state checks live together, woven in after each turn

```
@scenario(agent_fixture="agent", user_fixture="user")
async def test_billing_credit(s, store):
    (s.user_message("I was double charged in October")
     # O: the reply must insist on verifying first
     .assert_output(m.contains("verify"))
     # T: the agent must authenticate before account tools
     .assert_tool_calls([
         m.tool_call("authenticate_customer"),
         m.tool_call("get_billing", args={"id": "CUST-045"}),
     ])
     # S: a postcondition on the database / environment
     .assert_that(lambda: assert_credit_policy(store)))
```

- **O** `assert_output` — the reply
- **T** `assert_tool_calls` / `forbid_tool_calls` — the trace
- **S** `assert_that` — environment / DB state

One file reads like a dialogue with the contract inlined — no separate datasets, graders, and runner config.

One composable matcher language

- Literals, dicts and lists are coerced into matchers automatically.
- Rich combinators: `m.object`, `m.list`, `m.one_of`, `m.optional`, `m.tool_call(..., children=[...])`.
- Conditional rules:
`m.require("pager").when(m.field("severity") == "high")`.
- Selective **LLM rubrics** (`m.llm_criteria`) only where nuance is needed — pass/fail criteria, not opaque scores.

[FIGURE PLACEHOLDER]

Matcher cheat-sheet / nested `m.tool_call` example.
(Render a clean code card or reuse a report listing.)

Precise diagnostics: structured counterexamples

When a check fails, you learn which expectation broke, where

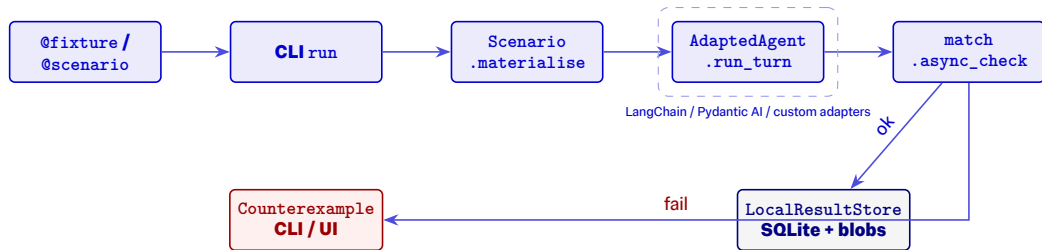
- Every failed assertion produces a **Counterexample**:
 - headline + human location
("assert_tool_calls after turn #2")
 - a **JSON-pointer path** to the mismatch, e.g. `$[1].args.line_id`
 - expected vs. actual, plus a bounded event trace
- Identical view in the **terminal**, the **SQLite** store, and the **web UI**.
- Turns “it failed somehow” into “this check broke here”.

```
progress: test_fail_arg_value [1/1]: queued
progress: test_fail_arg_value [1/1]: repeat started
FAIL test_fail_arg_value [1/1] 0.00s - assert_tool_calls: test_fail_arg_value: list item at index 1 did not match
the expected matcher for that position; re... FAIL test_fail_arg_value [1/1]

Check: assert_tool_calls
Where: 1st assert_tool_calls after turn #2 (agentturn) (tool call list for that turn)
Path: $[1].args.line_id
Expected:
equality: expected 'LINE-001'
[
  [0] tool_call("authenticate_customer"),
  [1] tool_call("order_replacement_sim", args=object({'line_id': 'LINE-001', ...}))
]
Actual:
[
  {
    "name": "authenticate_customer",
    "args": {
      "phone": "555-0100"
    },
    "result": "ok",
    "error": null,
    "children": [],
    "metadata": {}
  },
  {
    "name": "order_replacement_sim",
    "args": {
      "line_id": "LINE-002",
      "address_id": "ADDR-9"
    },
    "result": "ok",
    "error": null,
    "children": [],
    "metadata": {}
  }
]
list item at index 1 did not match the expected matcher for that position; representative mismatch at expected
index 1 (actual index 1): value does not equal expected
value does not equal expected
(+1 more matcher error(s))

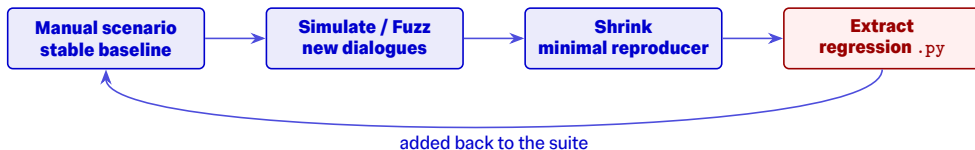
Event trace
└─ UserTurn: trace:auth-order
  └─ AgentTurn: completed 2 tool(s)
    └─ authenticate_customer result=ok
    └─ order_replacement_sim result=ok
```

Framework-neutral execution model



- Adapters **normalise traces** (LangGraph, Pydantic AI) into one event model — oracles never depend on a framework's message format.

From authored tests to automatic discovery



- **Repeats** for stochastic agents.
- **User simulation:** persona-driven multi-turn dialogues.
- **Mutation fuzzing:** perturb valid transcripts.

- **Shrinking:** delta-debug to a minimal failing dialogue.
- **Extraction:** pin a discovered failure as a permanent scenario.
- All under the same oracle vocabulary.

Persistent runs, CLI, and a web results browser

- Every run stored locally (SQLite + gzip blobs) with git metadata.
- CLI: run, runs, show, regressions, ui.
- Read-only **web UI**: runs list, run detail, trace drawer, fuzz trials.
- **Compare** named experiments side-by-side to track regressions.

run_20260526_152913_38e403 default • 26/05/2026, 16:29:13 • 24/05/2026, 16:48:19 • mean repeat 56.13s failed									
162 scenarios									
TRIALS	SCENARIO	TAGS	REPEAT	STATUS	DURATION	FAILURE KIND	FAILURE MESSAGE	ASSERTIONS	
—	test_007_c29_full	TELECOM FAULT-DETECTION FAULTY_FET TASK728 ORACLE_F MUTANT_FAULT_WRONG_ORACLE_CONFIGURATION	Q/1	failed	10.79s	forbid_test_calls	forbid_test_calls: test_007_c29_full: forbidden test call 'order_replacement_sen' found at index 0 (check...	Q/1	
—	test_007_c29_output	TELECOM FAULT-DETECTION FAULTY_FET TASK728 ORACLE_ID MUTANT_FAULT_WRONG_ORACLE_CONFIGURATION	1/1	passed	61.09s	—	—	Q/0	
—	test_007_c29_state	TELECOM FAULT-DETECTION FAULTY_FET TASK728 ORACLE_S MUTANT_FAULT_WRONG_ORACLE_CONFIGURATION	1/1	passed	33.97s	—	—	Q/0	
—	test_007_c29_trace	TELECOM FAULT-DETECTION FAULTY_FET TASK728 ORACLE_T MUTANT_FAULT_WRONG_ORACLE_CONFIGURATION	Q/1	failed	10.12s	forbid_test_calls	forbid_test_calls: test_007_c29_trace: forbidden test call 'order_replacement_sen' found at index 0 (check...	Q/1	
—	test_005_340_full	TELECOM FAULT-DETECTION FAULTY_FES TASK748 ORACLE_F MUTANT_FAULT_WRONG_NESTED_TOOL	Q/1	failed	1m 14s	assert_test_calls	assert_test_calls: test_005_340_full: could not match expected element at index 1 while preserving order...	Q/1	
—	test_005_340_output	TELECOM FAULT-DETECTION FAULTY_FES TASK748 ORACLE_ID MUTANT_FAULT_WRONG_NESTED_TOOL	1/1	passed	1m 57s	—	—	Q/0	
—	test_005_340_state	TELECOM FAULT-DETECTION FAULTY_FES TASK748 ORACLE_S MUTANT_FAULT_WRONG_NESTED_TOOL	1/1	passed	1m 2s	—	—	Q/0	
—	test_005_340_trace	TELECOM FAULT-DETECTION FAULTY_FES TASK748 ORACLE_T	Q/1	failed	1m 7s	assert_test_calls	assert_test_calls: test_005_340_trace: could not match expected element at index 1 while preserving...	Q/1	

Run detail page: per-scenario, per-repeat results with a trace drawer.

Live Demo

Writing a scenario, watching it fail precisely,
discovering a bug, and pinning it as a regression.

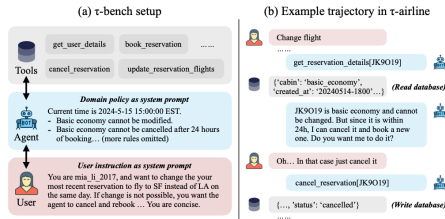
($\approx$ 8--10 minutes --- see speaker plan for the script)

Part II --- Evaluation

Does multi-surface, scenario-driven testing actually pay off?

TelcoSupportBench: the evaluation testbed

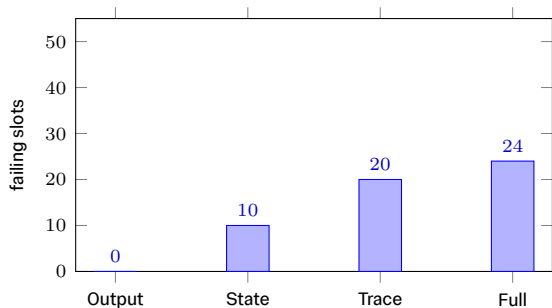
- A synthetic **mobile-support** domain built for this project.
- **50 tasks** (T01–T50) against a reference **LangGraph** agent: a coordinator + diagnostics + billing specialists.
- Each task has **four oracle lanes**: **O** output, **T** trace, **S** state, and **F** = full (all three).
- **Design principle**: tools stay permissive; **policy is asserted in the oracles**, not hidden in the simulator.
- Compared against **7** other evaluation approaches.



Inspiration: τ -bench-style dual-control support setting. [\[Replace with TelcoSupportBench architecture figure if available\]](#)

Finding 1 --- output-only grading is not enough

Reference agent across 200 oracle slots (50 tasks $\times$ 4 lanes)



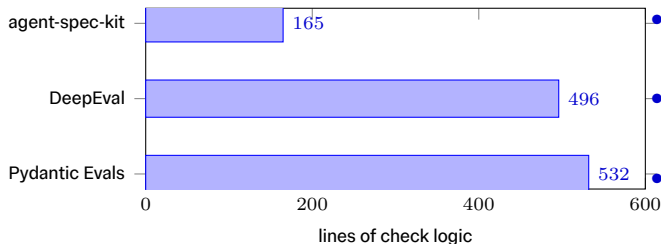
- **All 50 output lanes passed** — yet **54 of 200** slots failed.
- Every one of those failures lived in the **trace**, **state**, or **full** lanes.
- The reply “sounded correct” while a tool ran in the wrong order, an argument was wrong, or a store update never happened.

Takeaway

Final-answer evaluation alone would have reported **100%** pass.

Finding 2 --- more expressive, less code

Twelve canonical check patterns, ported across frameworks



- **165 lines** vs. 496 (DeepEval) and 532 (Pydantic Evals).
- **Highest clarity grade on every one** of the twelve specimens.
- On failures, it gives an exact **path witness** where alternatives return a scalar, broad prose, or a bare `AssertionError`.

Declarative DSL — equivalent in policy to the hand-written ports, but far easier to read and debug.

Finding 3 --- full oracles catch faults narrow lanes miss

Injected-fault detection (F01--F10) on previously-passing slots

- Ten fault families injected into the agent; measure which lane detects each.
- **Output** detection is frequently **0%** for tool-ordering and state-mutation faults.
- **State** and **Full** lanes reach **75–100%** on those same faults.
- The **combined oracle** is stronger than any single surface alone.

Fault	O	S	T	F
F02 (state)	0%	100%	67%	100%
F03 (state)	0%	100%	0%	100%
F05 (order)	0%	0%	100%	100%
F07 (trace)	50%	50%	100%	100%
F08 (trace)	0%	0%	75%	50%
F10 (full)	25%	0%	33%	100%

Detection rate = detected / eligible. Selected rows; full matrix F01–F10 in the report.

Finding 4 --- generative testing finds bugs nobody scripted

14 tasks, the same F-oracle throughout

Method	Convs	Tool paths	Failure sigs
Manual scripts	14	10	3
User simulation	70	58	24
Mutation fuzz	70	36	18

- Simulation & fuzzing reach **far more failure signatures** than the hand-written scripts — at little extra authoring cost.
- Discovered failures are **shrunk** and **extracted** into permanent regressions.

Honest about the limits

Threats to validity

- **Construct:** clarity & diagnostic grades used **LLM judges**, not a human developer study.
- **External:** a **single** reference agent and **one** telecom domain — design choices may favour the framework.
- **Rerun:** extraction produced a valid file for every eligible failure, but **fewer than half** reproduced the exact signature on immediate rerun — stochastic ordering/wording drifts.
- Results are **promising** but describe this setup, not agents in general.

Conclusion

- The right unit of evaluation is the **behavioural episode**, not the final reply — a polite answer can hide a skipped auth step, a mis-ordered tool, or a corrupted store.
- agent-spec-kit unifies **output + trace + state** oracles in one readable, executable, framework-neutral scenario.
- **Diagnostics are part of evaluation quality**: assertion-local counterexamples make failures fixable, not just countable.
- A single workflow spans **discovery** → **regression**: simulate, fuzz, shrink, extract.
- Evidence: hidden trace/state failures exposed, **3× less** check code with the highest clarity, and generative testing surfacing real bugs — e.g. a pre-auth privacy leak.

Future work

- More **domains and agent frameworks** — test whether the scenario model transfers beyond TelcoSupportBench.
- **Human studies** of authoring clarity and diagnostic usefulness, to replace LLM-judge grading.
- **Signature-stable replay** after extraction, and more robust matching of stochastic trajectories.
- **Larger, team-based suites** to see how the workflow scales in everyday development.

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Thank you.
Questions?

agent-spec-kit
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